COURSE MATERIAL	
SUBJECT	**ARTIFICIAL INTELLIGENCE(AM23APC301)**
UNIT	**3**
COURSE	**B.TECH**
DEPARTMENT	**CSE**
SEMESTER	**2-2**
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Version	**V-2**
PREPARED / REVISED DATE	**5-1-2025**

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1. Course Objectives

The objectives of this course is to

1. The students should be made to study the concepts of artificial intelligence.
2. The student should be made to learn the methods of solving problems using artificial intelligence.
3. The student should be made to introduce the concepts of Expert Systems.
4. To understand the applications of AI, namely game playing, theorem proving, and machine learning.
5. To learn different knowledge representation techniques.
6. To learn from experiences.

2. Prerequisites

Students should have knowledge on

1. concepts of artificial intelligence.

3. Syllabus

UNIT 3

REPRESENTATION OF KNOWLEDGE

Knowledge representation issues, Predicate logic- logic programming, Semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules-based deduction systems.

REASONING UNDER UNCERTAINTY

Review of probability, bayes probabilistic inferences, and Dempster-Shafer Theory

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4. Course outcomes:

1. Co1: Understand the AI, Intelligent agents and environment.
2. Co2: Apply the search Strategies and concept of problem reduction.
3. Co3: Understand the fundamentals of knowledge representation and reasoning
4. Co4: Apply the concepts of learning from observation.
5. C05: Analyze architecture of expert systems and language models.

5. Co-PO / PSO Mapping

Machine Tools	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3										2	2	3	2
CO2		3									2	2	3	2
CO3			3		2						3	3	3	3
CO4				3	2						3	3	3	3

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1		REPRESENTATION OF KNOWLEDGE	T2
2	1	INTRODUCTION	T2
3		Knowledge representation issues	T2
4		Predicate logic-logic programming	T2
5		Semantic nets-frames and inheritance	T2
6	2	Constraint propagation	T2
7		Representing knowledge using rules	T2
8		Rules-based deduction systems	T2
9		REASONING UNDER UNCERTAINTY	T2
10		Review of probability	T2
11	3	Bayes probabilistic inferences	T2
12		Dempster-Shafer theory	T2
13		REPRESENTATION OF KNOWLEDGE	T2

7. Activity Based Learning

1. Implementing the Natural Language Processing Model using Python

8. Lecture Notes

1. INTRODUCTION

1.1.1 Reinforcement learning might be considered to encompass all of AI: an agent is placed in an environment and must learn to behave successfully therein.

-To keep the chapter manageable, we will concentrate on simple environments and simple agent designs.

-For the most part, we will assume a fully observable environment, so that the current state is supplied by each percept.

- On the other hand, we will assume that the agent does not know how the environment works or what its actions do, and we will allow for probabilistic action outcomes.

Thus, the agent faces an unknown Markov decision process. We will consider three of the agents designs first introduced in Chapter 2:

• A utility-based agent learns a utility function on states and uses it to select actions that maximize the expected outcome utility.

Q-LEARNING • A Q-learning agent learns an action-utility function, or Q-function, giving the ex-Q-FUNCTION pelted utility of taking a given action in a given state.

• A reflex agent learns a policy that maps directly from states to actions.

A utility-based agent must also have a model of the environment in order to make decisions, because it must know the states to which its actions will lead.

For example, in order to make use of a backgammon evaluation function, a backgammon program must know what its legal moves are and how they affect the board position. Only in this way can it apply the utility function to the outcome states. A Q-learning agent, on the other hand, can compare the expected utilities for its available choices without needing to know their outcomes, so it does not need a model of the environment. On the other hand, because they do not know where their actions lead, Q-learning agents cannot look ahead.

Passive Learning, where the agent's policy is fixed and the task is to learn the utilities of states (or state–action pairs); this could also involve learning ACTIVE LEARNING a model of the environment. The principal issue is exploration: an agent must experience as much as possible of its environment in order to learn how to behave in it.

WHAT IS REPRESENTATION OF KNOWLEDGE:

Knowledge representation can be explained as the way artificial intelligence represents knowledge not with the help of stored data in the system but with prior experiences and knowledge to act like an intelligent human.

What are the 7 types of knowledge?

Explicit knowledge.
Implicit knowledge.
Tacit knowledge.
Procedural knowledge.
Declarative knowledge.

A Posteriori knowledge.
A Priori knowledge.

KNOWLEDGE REPRESENTATION ISSUES:

The fundamental goal of knowledge Representation is to facilitate inference (conclusions) from knowledge.

The issues that arise while using KR techniques are many. Some of these are explained below.

Important Attributed:

Any attribute of objects so basic that they occur in almost every problem domain? There are two attributed "instance" and "is a", that are general significance. These attributes are important because they support property inheritance.

Relationship among attributes:

Any important relationship that exists among object attributed?

The attributes we use to describe objects are themselves entities that we represent.

The relationship between the attributes of an object, independent of specific knowledge they encode, may hold properties like:

Inverse — This is about consistency check, while a value is added to one attribute. The entities are related to each other in many different ways.

Existence in an isa hierarchy — This is about generalization-specification, like, classes of objects and specialized subsets of those classes, there are attributes and specialization of attributes. For example, the attribute height is a specialization of general attribute physical-size which is, in turn, a specialization of physical-attribute. These generalization-specialization relationships are important for attributes because they support inheritance.

Technique for reasoning about values — This is about reasoning values of attributes not given explicitly. Several kinds of information are used in reasoning, like, height: must be in a unit of length, Age: of a person cannot be greater than the age of person's parents. The values are often specified when a knowledge base is created.

Single valued attributes — This is about a specific attribute that is guaranteed to take a unique value. For example, a baseball player can at time have only a single height and be a member of only one team. KR systems take different approaches to provide support for single valued attributes.

Choosing Granularity:

At what level of detail should the knowledge be represented?

Regardless of the KR formalism, it is necessary to know:

At what level should the knowledge be represented and what are the primitives?

Should there be a small number or should there be a large number of low-level primitives or High-level facts.

High-level facts may not be adequate for inference while Low-level primitives may require a lot of storage.

Example of Granularity:

Suppose we are interested in following facts:

John spotted SNIGDHA.

This could be represented as

Spotted (agent(John),object (SNIGDHA))

Such a representation would make it easy to answer questions such are:

Who spotted SNIGDHA?
Suppose we want to know:
Did John see SNIGDHA?
Given only one fact, we cannot discover that answer.
We can add other facts, such as
Spotted(x, y) -> saw(x, y)

Set of objects:

How should sets of objects be represented?
There are certain properties of objects that are true as member of a set but not as individual;
Example: Consider the assertion made in the sentences:
"there are more sheep than people in Australia", and
"English speakers can be found all over the world."
To describe these facts, the only way is to attach assertion to the sets representing people, sheep, and English.
The reason to represent sets of objects is: if a property is true for all or most elements of a set, then it is more efficient to associate it once with the set rather than to associate it explicitly with every elements of the set.
This is done,
in logical representation through the use of universal quantifier, and
in hierarchical structure where node represent sets and inheritance propagate set level assertion down to individual.
Finding Right structure:
Given a large amount of knowledge stored in a database, how can relevant parts are accessed when they are needed?
This is about access to right structure for describing a particular situation.
This requires, selecting an initial structure and then revising the choice.
While doing so, it is necessary to solve following problems:
How to perform an initial selection of the most appropriate structure.
How to fill in appropriate details from the current situations.
How to find a better structure if the one chosen initially turns out not to be appropriate.
What to do if none of the available structures is appropriate.
When to create and remember a new structure.
There is no good, general purpose method for solving all these problems. Some knowledge representation techniques solve some of these issues.

REPRESENTATION ISSUES:

Problem representation in artificial intelligence (AI) is the process of defining a problem in a way that an AI algorithm can understand and work with. It's important because an effective representation can make it easier for the AI to find a solution.
Here are some aspects of problem representation in AI:
Problem space
The problem space is the set of all possible states, actions, and transitions that can be encountered while solving a problem.
Problem formulation
This is the process of deciding what actions and states to consider, given a goal.
Search

This is the process of looking for a sequence of actions that reaches the goal.
Execution

This is the phase where the actions recommended by the solution are carried out. An effective problem representation defines the state space's boundaries and the transitions between states. A poorly defined problem representation can make it difficult for the AI to navigate the state space.

PREDICATE LOGIC AND LOGIC PROGRAMMING:

The Predicate Logic makes the representation of complex relationships simpler by representing them using the symbols like Predicates and Variables. Predicate Logic is used to represent many complex relationships. It can represent negation, conjunction, disjunction, and many more types of statements.

First-Order Logic in Artificial Intelligence

First-order logic (FOL), also known as predicate logic or first-order predicate calculus, is a powerful framework used in various fields such as mathematics, philosophy, linguistics, and computer science. In artificial intelligence (AI), FOL plays a crucial role in knowledge representation, automated reasoning, and natural language processing.

This article delves into the fundamentals of first-order logic, its components, and its applications in AI, providing a comprehensive overview of its significance and functionality.

Fundamentals of First-Order Logic

First-order logic extends **propositional logic** by incorporating quantifiers and predicates, allowing for more expressive statements about the world. The key components of FOL include constants, variables, predicates, functions, quantifiers, and logical connectives.

Constants: Constants represent specific objects within the domain of discourse. For example, in a given domain, **Alice**, **2**, and **NewYork** could be constants.

Variables: Variables stand for unspecified objects in the domain. Commonly used symbols for variables include **x**, **y**, and **z**.

Predicates: Predicates are functions that return true or false, representing properties of objects or relationships between them. For example, **Likes(Alice, Bob)** indicates that Alice likes Bob, and **GreaterThan(x, 2)** means that **x** is greater than 2.

Functions: Functions map objects to other objects. For instance, **MotherOf(x)** might denote the mother of **x**.

Quantifiers: Quantifiers specify the scope of variables. The two main quantifiers are:

Universal Quantifier (∀): Indicates that a predicate applies to all elements in the domain.

For example, $\forall x$ (Person(x) $\rightarrow$ Mortal(x)) means "All persons are mortal."

Existential Quantifier ($\exists$): Indicates that there is at least one element in the domain for which the predicate holds.

For example, $\exists x$ (Person(x) $\wedge$ Likes(x, IceCream)) means "There exists a person who likes ice cream."

Logical Connectives: Logical connectives include conjunction ($\wedge$), disjunction ($\vee$), implication ($\rightarrow$), biconditional ($\leftrightarrow$), and negation ($\neg$). These connectives are used to form complex logical statements.

Syntax and Semantics of First-Order Logic.

The syntax of first-order logic specifies the rules for constructing valid expressions, including terms and formulas. Terms are expressions that refer to objects and include constants, variables, and functions. Formulas are logical statements formed by combining predicates, terms, quantifiers, and logical connectives.

The semantics of FOL define the meaning of the expressions. An interpretation assigns a domain of discourse and provides meaning to the constants, functions, and predicates. A formula is considered true under an interpretation if it accurately describes the relationships and properties of the objects in the domain.

For instance, in the domain of natural numbers, let GreaterThan(x, y) be a predicate that holds if x is greater than y. If x is assigned the value 5 and y is assigned the value 3, then GreaterThan(5, 3) would be true under this interpretation.

Applications of First-Order Logic in AI

First-order logic is instrumental in various AI applications, enabling systems to reason about complex knowledge structures and make informed decisions. Some of the key applications of FOL in AI include:

Knowledge Representation: FOL provides a robust framework for representing complex relationships and properties of objects. For example, in a medical diagnosis system, predicates can represent symptoms, diseases, and their relationships, allowing the system to reason about potential diagnoses.

Automated Theorem Proving: Automated theorem proving involves using algorithms to prove mathematical theorems. FOL provides the foundational structure for many theorem proving systems, enabling them to verify software and hardware correctness, prove mathematical theorems, and more.

Natural Language Processing (NLP): In NLP, FOL is used to parse and understand natural language by representing the meaning of sentences in a formal, logical manner. This allows AI systems to perform tasks such as question answering, machine translation, and text summarization.

Expert Systems: Expert systems encode expert knowledge using FOL and reason about it to make decisions. For example, a legal expert system might use FOL to represent legal rules and infer outcomes based on given facts, aiding in legal decision-making processes.

Semantic Web: The Semantic Web uses FOL to describe relationships between web resources, enabling more intelligent information retrieval and processing. This enhances the ability of AI systems to understand and interact with web content.

Example: Using First-Order Logic for Reasoning

To illustrate the use of FOL in reasoning, consider the following knowledge base:

∀x (Cat(x) → Mammal(x)) (All cats are mammals)

∀x (Mammal(x) → Animal(x)) (All mammals are animals)

Cat(Tom) (Tom is a cat)

From these statements, we can infer:

Mammal(Tom) (Since Tom is a cat, and all cats are mammals)

Animal(Tom) (Since Tom is a mammal, and all mammals are animals)

SEMANTIC NETS –FRAMES AND INHERITANCE:

Semantic Network Representation Semantic networks are alternative of predicate logic for knowledge representation.

In Semantic networks, we can represent our knowledge in the form of graphical networks. This network consists of nodes representing objects and arcs which describe the relationship between those objects.

Semantic networks can categorize the object in different forms and can also link those objects. Semantic networks are easy to understand and can be easily extended. This representation consists of mainly two types of relations: IS-A relation (Inheritance) Kind-of-relation Example: Following are some statements which we need to represent in the form of nodes and ARCS

EXAMPLE:

Jerry is a cat.

Jerry is a mammal

Jerry is owned by Priya.

Jerry is brown colored.

All Mammals are animal.

Practice Quiz

1. What is the field of Natural Language Processing (NLP)?
a) Computer Science
b) Artificial Intelligence
c) Linguistics
d) All of the mentioned

2. Which of the following is the main challenge of NATURAL LANGUAGE PROCESSING?

(A). Handling Ambiguity of Sentences

(B). Handling Tokenization

(C). Handling POS-Tagging

(D). All of these

3. NLP is concerned with the interactions between computers and human (natural) languages.
(a) TRUE
(b) FALSE

4. What is Coreference Resolution?
(a) Anaphora Resolution
(b) Given a sentence or larger chunk of text, determine which words ("mentions") refer to the same objects ("entities")
(c) All of the mentioned
(d) None of the mentioned

5. What is Machine Translation?
(a) Converts one human language to another
(b) Converts human language to machine language
(c) Converts any human language to English
(d) Converts Machine language to human language

6. _____is an area of Machine Learning in which about taking suitable action to maximize reward in a particular situation.

A) Supervised learning
B) unsupervised learning
C) Reinforcement learning
D) None of these

7. _____Reinforcement is defined as when an event, occurs due to a particular behavior.

A) negetive
B) positive
C) neutral
D) None of these

8. There are _____types of reinforcement.

A) 3
B) 2
C) 4
D) None of these

9. Which of the following is not Advantages of reinforcement learning?

A) Maximizes Performance
B) Sustain Change for a long period of time
C) Too much Reinforcement can lead to overload of states which can diminish the results
D) None of these

10. Reinforcement learning is one of _____basic machine learning paradigms

A) 5
B) 4
C) 2
D) 3

9. Assignments

S.No	Question	BL	CO
1	Explain about Active Reinforcement Learning and Passive Reinforcement Learning	2	1
2	Write about Generalization in Reinforcement Learning.	2	1
3	Write about Applications of RL.	2	1
4	Explain the types of Language Models in Natural Language Processing	2	1
5	Describe the Text Classifications.	3	1

10. Part A- Question & Answers

S.No	Question& Answers	BL	CO

1	1. What is Reinforcement Learning? 1. A utility-based agent learns a utility function on states and uses it to select actions that maximize the 2. expected outcome utility. 3. A Q-learning agent learns an action-utility function, or Q-function, giving the expected utility 4. of taking a given action in a given state. 5. A reflex agent learns a policy that maps directly from states to actions.	**1**	**1**
2	2. Define Active Reinforcement Learning Ans) Active reinforcement learning (ARL) is **a variant on reinforcement learning where the agent does not observe the reward unless it chooses to pay a query** cost c > 0. The central question of ARL is how to quantify the long-term value of reward information.	**1**	**1**
3	3) . Define Passive Reinforcement Learning Ans) In case of passive RL, **the agent's policy is fixed which means that it is told what to do**. ... Therefore, the goal of a passive RL agent is to execute a fixed policy (sequence of actions) and evaluate it while that of an active RL agent is to act and learn an optimal policy.	**2**	**2**
4	4) What is Natural Language Processing? Ans) Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that **enables machines to understand the human language**. Its goal is to build systems that can make sense of text and automatically perform tasks like translation, spell check, or topic classification	**1**	**3**
5	**5)** . Define Information Retrieval. **Ans) the techniques of storing and recovering and often disseminating recorded data especially through the use of a computerized system**	**1**	**1**

6	6)Define Information Extraction. Ans) Information extraction (IE) is the task of **automatically extracting structured information from unstructured and/**or semi-structured machine-readable documents, while information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information.	1	1

11. Part B- Important Questions

S.No	Question	BL	CO
1	**Explain about Active Reinforcement Learning and Passive Reinforcement Learning**	1	1
2	**Write about Generalization in Reinforcement Learning.**	2	1
3	**Write about Applications of RL.**	2	1
4	**Explain the types of Language Models in Natural Language Processing**	2	1
5	**Describe the Text Classifications.**	3	1
6	**Write about Information Retrieval and Information Extraction.**		

12. Supportive Online Certification Courses

1. Opportunities in Artificial Intelligence and Enabling Technologies for Internet of Things(IoT) FDP by AICTE – 2 weeks
2. Artificial Intelligence by Python organized by Brain Vision Solutions Pvt Limited– 1 week.

13. Real Time Applications

S.No	Application	CO
1	**AI in space exploration.**	3
2	**AI used in self driving vehicles**	2
3	**AI used in gamming**	1
4	**AI used in space exploration**	3

14. Prescribed Text Books & Reference Books , Text Book:

Textbooks:

1. Stuart J.Russell, Peter Norvig, "Artificial Intelligence A Modern Approach", 3rd Edition, Pearson Education, 2019.

Reference Books:

1. Nilsson, Nils J., and Nils Johan Nilsson. Artificial intelligence: a new synthesis. Morgan Kaufmann, 1998.
2. Johnson, Benny G., Fred Phillips, and Linda G. Chase. "An intelligent tutoring system for the accounting cycle: Enhancing textbook homework with artificial intelligence." Journal of Accounting Education 27.1 (2009): 3039.

17. **MINI PROJECT SUGGESTION:**

1. Student Information Chatbot Project:

Design CartPole Game using OpenAI Also known as the "Inverted Pendulum" problem, CartPole is a reinforcement game provided by OpenAI Gym. In this game, there is a cart to which a pole is attached which is unstable as the centre of mass of the pole is above the pivot point. The player is supposed to move the cart left or right to balance it. If the pole deviates more than 15 degrees, the game ends. Otherwise, for every timestamp e